Syllabus: Machine Learning in Economics

Course & Instructor Information

Course

Course Title, Prefix, Number, Section: Machine Learning in Economics, ECON 40333-045

Semester and Year: Fall 2026

Number of Credits: 3

Course Component Type: LEC

Class Location: SCHAR 2010

Class Meeting Day(s) & Time(s): TR 12:30-1:50 PM

Instructor

Instructor Name: Stepan Gordeev

Office Location: SCHAR 4108

Office Hours: M 4:30-5:30 PM, R 3:30-4:30 PM

Preferred Method of Contact: Email

Email: s.gordeev@tcu.edu

Response Time: 24 hours on weekdays

Final Evaluative Exercise

Final Project Report Workshop
December 15, 11 AM - 1:30 PM
write a report analyzing final project

Note for students: The syllabus is your first course reading. It provides an orientation to, overview of the flow, and expectations of the course. You should turn to the syllabus for details on assignments and course policies.

Student Resources & Policy Information

Click or scan QR code for resources to support you as a TCU student. Please note section on Student Access and Accommodation and Academic Conduct & Course Materials Policies.

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Course Description

Catalog Description

Econometric and machine learning methods for dealing with big data and non-standard data. Introduction to programming in R, linear regression, variable selection, random forests, unsupervised learning, deep learning.

Section Description

The quantity and complexity of data available to economists grows every year. This course will introduce you to the methods of analyzing high-dimensional data that conventional statistical and econometric techniques are not well-suited for. The methods we will cover include linear regression, variable selection, trees, random forests, clustering, neural networks, and large language models. In the process of learning these methods, you will also become proficient in R: a popular programming language used for statistical and econometric analysis.

Prerequisites

MATH 10043 or MATH 30853 or INSC 20153.

Course Materials

Recommended Materials

James et al., *An Introduction to Statistical Learning with Applications in R*. ISBN 978-1071614174. Abbreviated as ISL below. You can download a free digital copy from the authors' website: https://www.statlearning.com/

Learning Outcomes

Course Learning Outcomes

At the conclusion of the course, students will be able to:

1. choose an appropriate econometric or machine learning method for analyzing the data at hand;
2. implement the method in R;
3. estimate the economic quantities of interest using the method;
4. assess the successes and the limitations of the chosen method;
5. describe and interpret the analysis in a compelling way.

COURSE REQUIREMENTS

Instructions for assignments will be distributed as .qmd files (Quarto notebooks). Each assignment below is to be submitted as a .qmd file with your work and a "rendered" .html output to the Box folder that will be shared with you, in the sub-folder corresponding to the assignment. Make sure that each .qmd file is fully reproducible (I should be able to run the file and produce the same results). Grades and feedback will be delivered via TCU Online.

In-Class Exercises

In most classes, you will be given an exercise to complete before the end of the same class. You must bring a laptop to class to complete these exercises. You will work on these in small groups, but each student must make their own submission. You cannot use external resources (other than official documentation) or AI assistance for these exercises. Two lowest scores will be dropped. Course learning outcomes covered: 2-3.

Assignments

You will be given five bigger assignments to complete outside of class. These will build and expand on in-class exercises. These assignments must be completed individually. One lowest score will be dropped. Course learning outcomes covered: 1-4.

Milestone Projects

The assignments will be grouped into two modules devoted to particular sets of methods. Assignments within a single module will build on top of one another, guiding you through completing a significant big data project. You will need to compile your assignment submissions belonging to a single module into a single project, expanding your analysis when requested and ensuring that the shortcomings of individual assignment submissions have been addressed. Each module's project will represent the milestone associated with that module. Course learning outcomes covered: 5.

Final Project

You will develop and conduct a data analysis project using a dataset of your choice (from a list of suggestions).

First, you will prepare a proposal stating the dataset you chose, the question you plan to answer, and the methods you intend to use.

Next, you will conduct the work outlined in your proposal. You will submit a brief Final Project Progress Report outlining the initial steps you complete.

Finally, you will prepare a Final Project submission following the format of the Milestone Projects, but rendered as a PDF. The Final Project will showcase your proficiency in econometric and machine learning methods for working with big data. You will submit the Final Project by the due date.

The final evaluative exercise is an in-person Final Project Report Workshop. You will receive a printed copy of your final project submission, your original proposal, an AI-generated baseline sample based on your proposal, and a set of questions that will

guide you through analyzing the strengths and limitations of your final project, how it evolved from your proposal, and how your work exceeds an AI-generated baseline. You will answer these questions in a handwritten report and submit it.

Grading Policy

Late Work

No late work will be accepted, unless mandated by the Official TCU Absence Policy.

Participation, Engagement & Attendance

There is no explicit grading for attendance and participation. At the same time, if you are not present in class on a particular day, you will not be able to complete the in-class exercise.

Absences described in the Official University Absence Policy will be excused. Excused Absences or Official University Absences are absences described in the Official University Absence Policy and include the following: Title IX related issues, military leave, holy days, and university related absences.

Verified Absences may be excused following the instructor's discretion. Because it is considered an infringement on student privacy for me to have access to student medical records, I cannot accept medical documentation to justify absences. If you have a legitimate reason for your absence and want to provide verification, please access the Absence Documentation Form here.

Course Assignments & Final Grade

Category	Percentage
In-Class Exercises	20
Assignments	30
Milestone Projects	20
Final Project	
Proposal	5
Progress Report	5
Final Project	15
Workshop Report	5
Total	100

Grading Scale

Grade	Percentage
A	94–100
A-	90–93.99
B+	87–89.99
B	84–86.99
B-	80–83.99
C+	77–79.99
C	74–76.99
C-	70–73.99
D+	67–69.99
D	64–66.99
D-	60–63.99
F	0–59.99

Course Policies

Technology Policies

Please turn off or silence cell phones and put them away during class. You may use laptops for the in-class exercises but *not during the lectures*.

Artificial intelligence (AI) may not be used for in-class exercises and workshop report. It may be used for all other types of assignments. If you are in doubt as to whether you are using AI appropriately in this course, it is your responsibility to discuss your situation with the instructor.

COURSE SCHEDULE

The calendar represents a tentative schedule. Plans may need to be changed as the semester unfolds.

Dates	Topic	Assigned Materials	Due
Week 1: Aug 25, 27	Syllabus, Intro, R	Read Syllabus	
Week 2: Sep 1, 3	Basics of Programming in R	ISL 2	
Week 3: Sep 8, 10	Reproducible Research		
Week 4: Sep 15, 17	Data Visualization		
Week 5: Sep 22, 24	Linear Regression	ISL 3.1	Assignment 1
Week 6: Sep 29, Oct 1	Polynomial Regression	ISL 3.2	
Week 7: Oct 6, 8	Logistic Regression	ISL 4.3	Assignment 2
Week 8: Oct 13, 15	Evaluating Performance	ISL 5	
Week 9: Oct 20, 22	Variable Selection	ISL 6	Project 1
Week 10: Oct 27, 29	Trees, Random Forests	ISL 8.1, 8.2	Assignment 3
Week 11: Nov 3, 5	Tuning, Clustering	ISL 12.4	Assignment 4
Week 12: Nov 10, 12	Neural Networks	ISL 10.1, 10.2	Proposal
Week 13: Nov 17, 19	Transformers		Project 2
Week 14: Nov 24, 26	Thanksgiving Break		
Week 15: Dec 1, 3	Text as Data		Progress Report
Week 16: Dec 8	AI		Final Project
Finals Week: Dec 15	Final Project Workshop		